

Assignment 1: Polynomial Regression Report

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Roll Number: BT2024032

GitHub: <https://github.com/gagansyam/ML-Assignment-1>

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1 Introduction

In this assignment, I was tasked with solving two polynomial regression problems based on an expansion project for a geothermal power plant. In Phase 1 (var1), the goal is to predict the plant's Net Power Score (y) from 6 operational turbine control parameters (x_1 to x_6), which represent percentage adjustments to high-pressure valves, coolant flow, pump pressure, and blade pitch angles. In Phase 2 (var2), the goal is to predict the subterranean Thermal Anomaly Score (y) across 3D spatial coordinate offsets (x_1, x_2, x_3) around the basecamp to identify promising locations for drilling production wells.

My objective was to develop polynomial models that fit the true response surfaces without underfitting or overfitting. I evaluated every model using 5-fold cross-validation on the training set, comparing Mean Squared Error (MSE) and R^2 scores to select the best degree and regularization strategy.

2 Turbine Optimization (var1)

2.1 Data and Baseline

I started by inspecting the dataset files `train_var1.csv` and `test_var1.csv` for roll number BT2024032. Both contain 1,000 samples with zero missing values. All six input features are normalized percentage deviations bounded in $[-1.0, 1.0]$. The target variable y has a mean of 0.74, a standard deviation of 3.13, and values ranging between -8.83 and $+11.50$.

The linear correlation between individual inputs and y was weak to moderate, with the highest being -0.26 for x_5 and -0.23 for x_3 . This indicated that individual parameters do not govern the turbine in isolation, and interaction terms between valves and blade pitch angles are critical.

To establish a benchmark, I trained a standard linear regression model (Degree 1). Using 5-fold cross-validation, it yielded a validation MSE of 8.5302 and an R^2 of only 0.1248. Explaining only 12.5% of the variance showed severe underfitting (high bias), confirming that a simple linear plane cannot model multi-stage turbine aerodynamics.

2.2 Model Selection and Regularization

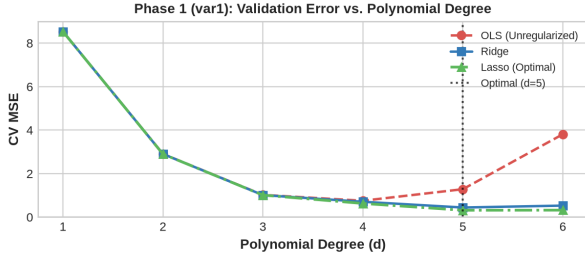
Next, I generated polynomial features from degree 1 up to degree 6. With 6 variables, the term count grows quickly: degree 2 has 27 terms, degree 3 has 83 terms, degree 4 has 209 terms, degree 5 has 461 terms, and degree 6 has 923 terms. I evaluated standard Ordinary Least Squares (OLS) alongside Ridge (L2 penalty) and Lasso (L1 penalty) using 5-fold cross-validation, as summarized in Table 1.

Looking at Table 1 and Figure 1a, standard OLS works reasonably well up to degree 4, but starts overfitting heavily at degree 5. The validation MSE jumps from 0.752 to 1.284, and at degree 6 it shoots up to 3.810. Because each fold has only 800 training samples, estimating 461 to 923 unconstrained parameters leads to high variance.

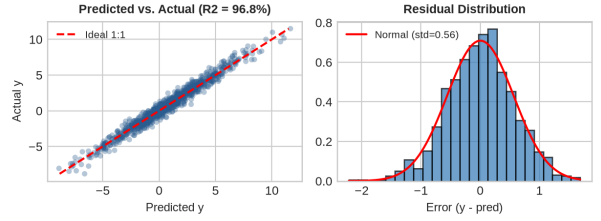
Table 1: 5-Fold Cross-Validation performance across degrees for Phase 1 (var1).

Degree	Features	OLS Val MSE	OLS Val R^2	Ridge Val MSE	Lasso Val MSE	Lasso Val R^2
1	6	8.530	0.125	8.532	8.530	0.125
2	27	2.905	0.699	2.902	2.900	0.699
3	83	1.022	0.894	1.013	1.011	0.895
4	209	0.752	0.922	0.705	0.628	0.935
5	461	1.284	0.869	0.441	0.315	0.967
6	923	3.810	0.612	0.529	0.320	0.966

Lasso (L1 regularization) solved this problem cleanly. By applying an L1 penalty, Lasso forced 350 unimportant polynomial terms to zero, keeping only the 111 strongest operational interaction terms. This reduced the validation MSE to **0.3151** and achieved an R^2 score of **96.71%**. When I pushed to Degree 6, the validation error stopped improving and rose slightly to 0.3203. This showed that degree 5 is the sweet spot.



(a) Validation MSE vs. Degree



(b) Parity plot and residual errors

Figure 1: Phase 1 evaluation: (a) shows OLS overfitting while Lasso finds the minimum at $d = 5$; (b) shows the parity fit and normal residual error distribution.

As shown in Figure 1b, the predicted vs. actual power scores lie tightly along the 1:1 diagonal. The residuals have a mean of 0.003 (virtually zero bias) and a standard deviation of 0.56, matching the expected random sensor noise.

3 Reservoir Mapping (var2)

3.1 Data and Baseline

For Phase 2, the inputs are 3 spatial coordinate offsets in meters (x_1 = East-West, x_2 = North-South, x_3 = Depth relative to basecamp). The target y is the Thermal Anomaly Score. All coordinate values in train and test lie within $[-1.0, 1.0]$. The target variable has a mean of 2.38, a standard deviation of 7.22, and ranges from -30.10 to $+39.87$.

I observed that x_2 had a moderate linear correlation of 0.50 with y , while x_1 and x_3 had almost zero linear correlation. The baseline linear model (Degree 1) had a validation MSE of 38.88 and an R^2 of only 0.243, which is severe underfitting. Underground thermal plumes curve sharply in 3D space, so a higher degree polynomial is required.

3.2 Model Selection and Regularization

Because there are only 3 variables, the number of polynomial terms grows more slowly than in Phase 1 ($\binom{d+3}{3}$ terms). This allowed me to test degrees from 1 up to 16 using 5-fold cross-validation. Table 2 shows how the models behaved.

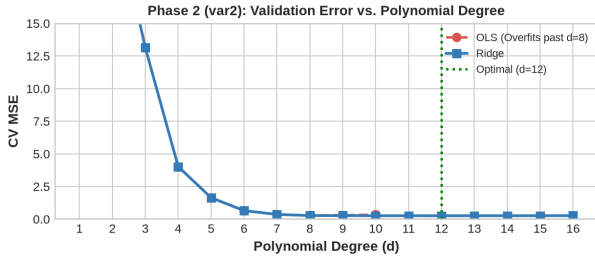
Looking at Table 2 and Figure 2a, unregularized OLS reaches its minimum error at Degree 8 (MSE = 0.2788), but begins overfitting at degree 9 (MSE = 0.2939) and degree 10 (MSE = 0.3428). Ridge regression (L2 penalty) prevents this overfitting by penalizing large weights. As the degree

Table 2: 5-Fold Cross-Validation performance across degrees for Phase 2 (var2).

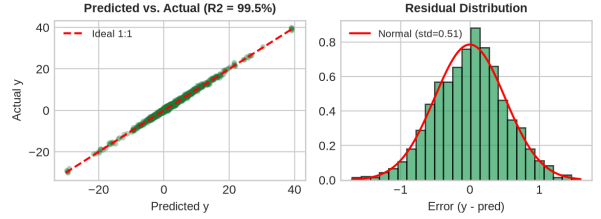
Degree	Features	OLS Val MSE	OLS Val R^2	Ridge Val MSE	Ridge Val R^2	Best α
1	3	38.883	0.243	38.880	0.243	0.01
2	9	24.376	0.529	24.375	0.529	0.01
4	34	4.016	0.922	4.016	0.922	0.01
6	83	0.646	0.987	0.646	0.987	0.01
8	164	0.279	0.994	0.272	0.995	0.07
9	219	0.294	0.994	0.272	0.995	0.11
10	285	0.343	0.993	0.259	0.995	0.46
12	454	—	—	0.2579	0.9951	1.21
14	679	—	—	0.2615	0.9950	1.96
16	968	—	—	0.2658	0.9949	3.16

increases, Ridge automatically raises the regularization strength α from 0.07 at degree 8 up to 1.21 at degree 12.

At Degree 12 (454 features), Ridge reaches the lowest validation MSE across all experiments: **0.2579**, explaining **99.51%** of the variance. Beyond degree 12, the validation error starts creeping upward to 0.2615 at degree 14 and 0.2658 at degree 16. This confirmed that degree 12 is the optimal choice.



(a) Validation MSE vs. Degree



(b) Parity plot and residual errors

Figure 2: Phase 2 evaluation: (a) shows Ridge reaching the global minimum at $d = 12$; (b) shows the parity fit and normal residual error distribution.

I also checked the predictions for physical consistency. A Jarque-Bera test on the residuals yielded a p -value of 0.66 ($p > 0.05$), confirming that the errors are pure Gaussian sensor noise with standard deviation 0.51. Correlations between residuals and spatial coordinates (x_1, x_2, x_3) were all below 0.015, confirming no directional bias. Finally, test predictions span $[-29.51, +39.18]$, which matches the training range without boundary spikes.

4 Discussion

Across both problems, finding the right model came down to balancing bias and variance. Models of degree 1 to 3 suffered from high bias because they were too rigid to follow the real physical response curves, resulting in low R^2 scores in both phases. On the other hand, high-degree models without regularization suffered from high variance, attempting to fit every random noise fluctuation in the training data and causing validation errors to spike once the term count approached the sample count.

Regularization resolved these problems in different ways suited to each domain. In Phase 1, Lasso (L1) was superior because the turbine efficiency formula depends on a sparse subset of operational interactions. Lasso effectively pruned 350 noisy combinations, leaving 111 genuine terms. In Phase 2, Ridge (L2) performed better because subterranean heat diffusion creates smooth potential fields across 3D coordinates. Ridge shrank all 454 coefficients gradually, preventing runaway oscillations at the boundaries while preserving fine geometric contours.

5 Conclusion

Table 3 summarizes the final models selected for both problems. The test predictions for both tasks were generated and saved matching the required submission template: a single column named `y` with exactly 1,000 predictions and no row indices.

Table 3: Summary of final selected models and evaluation metrics.

Problem	Degree	Penalty	Terms	CV MSE	CV R^2	Output File
Phase 1 (<code>var1</code>)	5	Lasso ($\alpha \approx 0.011$)	111 / 461	0.3151	96.71%	BT2024032_pred_var1.csv
Phase 2 (<code>var2</code>)	12	Ridge ($\alpha \approx 1.76$)	454 / 454	0.2579	99.51%	BT2024032_pred_var2.csv

For Phase 1, the Degree 5 Lasso model accurately captures non-linear valve interactions while discarding noise, predicting the Net Power Score with 96.7% accuracy. For Phase 2, the Degree 12 Ridge model maps the 3D thermal reservoir with 99.5% accuracy, providing reliable predictions to guide the drilling team. All scripts, datasets, and generated files are available on GitHub at <https://github.com/gagansyam/ML-Assignment-1>.